

Memo 03: MP1 Mass-Conservation Check and Diagnostic Hypotheses

FVCOM-MP Delaware Test Cases

April 28, 2026

Purpose

This memo summarizes the latest mass-budget extraction for experiments **a**, **b**, and **c**, and lists concise diagnostic hypotheses for the apparent mass loss and oscillation in the floc-enabled cases.

Current Evidence

The MATLAB extractors integrate water-column and bed MP1 mass from the NetCDF output files. The Python script `FVCOM_MP_test_run/PYTHON/analyze_mp1_mass_conservation.py` then compares the MAT files using a common absolute time axis. Case **a** has now been rerun for the full 10-day window with output beginning at the same model day as cases **b/c**; all three cases start at FVCOM day 58485.0.

Case	Records	Time window (comparison days)	Initial total (kg)	Final total (kg)	Final bed (kg)
a	240	0.00–9.96	0.00	1001.25	0.00
b	240	0.00–9.96	0.00	154.71	0.00
c	240	0.00–9.96	0.00	183.58	0.58

The common comparison window is now day 0.00–9.96. At the shared endpoint, the retained total masses are approximately

$$M_a = 1001.25 \text{ kg}, \quad M_b = 154.71 \text{ kg}, \quad M_c = 183.58 \text{ kg}.$$

Thus case **b** retains about 15.5% of case-**a** mass and case **c** retains about 18.3%. The corrected full-window comparison confirms that the large deficit in the floc-enabled cases is not an artifact of the previous case-**a** output offset.

Important Consistency Check

For floc cases, the split variables are internally consistent at output:

$$\text{mp1_agg} + \text{mp1_dis} \approx \text{mp1}.$$

The maximum mismatch is zero for case **b** and approximately 1.35×10^{-7} kg for case **c**. Therefore, the saved-output recombination of `conc_a` and `conc_d` is not the leading suspect.

Source-Path Check

`plast(iplast)%cdis` is the river-source concentration path, not the open-boundary flux workspace. It is filled from the river NetCDF variable `mp1(time,rivers)` and paired with `river_flux(time,rivers)` inside `adv_scal`. For the present files, `mp1` is in kg m^{-3} , and the cases use `RIVER_TS.SETTING='calculated'`, so `bcond_scal.PTsource` does not add a second source. With `SOURCE_AG_RATE=0`, the flocc-mode river input is

$$QC = QC(1 - \text{SOURCE_AG_RATE})$$

into `conc_d`, and zero into `conc_a`. Therefore, the present working assumption is that the imposed river mass source is conservative.

Diagnostic Hypotheses

1. **Null-floc split path is not guaranteed identical to no-floc path.** The most focused question is now: when the aggregation/disaggregation kinetic terms are disabled and `SOURCE_AG_RATE=0`, why does the floc `conc_d` pathway not reproduce the no-floc `conc` pathway? This comparison should be exact if every operator applied to `conc_d` is numerically identical to the operator applied to `conc`.
2. **Final exchange/recombination ordering was asymmetric.** The split path previously clipped `conc_a/conc_d` but then exchanged `cnew_a/cnew_d`; the no-floc path exchanged the clipped combined `conc`. This could leave exchanged halo state and prognostic split state inconsistent. The code has been patched so the split path now clips `cnew_a/cnew_d`, exchanges those same arrays, then sets

$$\text{conc} = \text{cnew} = \text{conc_a} + \text{conc_d}.$$

3. **Floc-only clips and caps can act as silent sinks.** The floc path applies upper caps to `cnew_a/cnew_d` and nonnegative clipping after transport. If split advection creates undershoots, the limiter removes mass unless the removed amount is tracked. This remains a high-priority checkpoint for a null-floc comparison.
4. **Erosion and surface-flux update are not algebraically identical.** In the floc branch, erosion and surface inputs are split by `ratio_bottom/ratio_surface` and are subject to a per-step “10x” growth limiter. In the no-floc branch, `eflx` and `pflx` are added directly to `conc`. This is a concrete asymmetry even when kinetics are disabled.
5. **Bed exchange is unlikely to explain the large loss in this run.** Case `b` has essentially no bed mass, and case `c` has only about 0.58 kg bed mass at the end. Deposition/erosion is therefore not large enough by itself to explain the 817–847 kg difference relative to case `a`.
6. **Open-boundary flux storage is lower priority for the null test.** `cflx` is an open-boundary flux workspace, not the river source. It is still worth logging, but with `OBC_AG_RATE=0` and all source mass entering `conc_d`, it is not currently the leading explanation for why `conc_d` should fail to match `conc` in the null-kinetics test.

Asymmetry Checklist for `conc` vs. `conc_d`

For the next debugging pass, step through these operators in order and compare domain-integrated mass and field differences after each one:

1. **Initialization and river input.** With `INI_CONC_AG_RATE=0`, `INI_MASS_AG_RATE=0`, and `SOURCE_AG_RATE=0`, initial and river-source mass should enter only `conc_d`; the source integral should match no-floc $\sum QC$.
2. **Settling/deposition.** In the null-floc limit, `conc_d` uses `Calc.WsetPlastic` and `settle_flux_new`, matching the no-floc `conc` branch. Confirm identical `wset`, `dflx`, and water-column mass change.
3. **Erosion/surface update.** Compare direct no-floc addition against the split branch's ratio logic and 10x limiter. This is currently a leading candidate for non-equivalence if erosion or surface flux is active.
4. **Horizontal advection and vertical diffusion.** Compare `adv_scal` and `vdif_scal` outputs for `conc` and `conc_d`. In null-floc mode, these should match to roundoff.
5. **Kinetics.** Force `lambda_a=lambda_d=0` or bypass the kinetic update, then verify that this block is exactly conservative and makes no change to `conc_d`.
6. **Caps, nonnegative clips, and MPI exchange.** Track mass removed by hard caps and nonnegative clipping. After exchange, verify `conc = conc_a + conc_d` and `mass = mass_a + mass_d`.

Focused Next Test

1. Run a dedicated null-floc case with `PLAST_FLOC=T`, `SOURCE_AG_RATE=0`, `INI_CONC_AG_RATE=0`, and `lambda_a=lambda_d=0`. Compare `mp1_dis` directly against case-a `mp1`.
2. First run this comparison on one processor. If it matches in serial but not in parallel, the remaining issue is likely an exchange/halo-state ordering problem.
3. If serial already diverges, walk through the asymmetry checklist above, starting with settling/deposition and erosion/surface update.